Later on in the book, we sometimes show isolated snippets from a larger project; these are *not* meant for you to run on your computer. If you're interested in running them on your own, you can download all the project files from the book's source code repository.

Most chapters have a "Your Turn" section with exercises for you to try. Don't skip these! Practicing on your own will ensure that each chapter builds on the skills you've honed over the course of the book.

RSpec and Behavior-Driven Development

RSpec bills itself as a behavior-driven development (BDD) test framework. We'd like to take a moment to talk about our use of that term, along with a related term, test-driven development (TDD).

Without TDD, you might check your program's behavior by running it manually or by writing a one-off test harness. In situations where you intend to scrap the program shortly afterward, these approaches are all right. But when long-term maintenance is a priority, TDD provides important benefits.

With TDD, you write each test case just before implementing the next bit of behavior. When you have well-written tests, you wind up with more maintainable code. You can make changes with the confidence that your test suite will let you know if you've broken something.

The term TDD is a bit of a misnomer, though. Despite the fact that it has the word "test" in the name, TDD isn't just about your tests. It's about the way they enable fearless improvements to your design. For this reason, Dan North coined the term behavior-driven development in 2006 to encapsulate the most important parts of TDD.[1]

BDD brings the emphasis to where it's supposed to be: your code's *behavior*. The community stresses the importance of expressiveness in your tests, something that we'll be talking about a lot in this book. BDD is also about treating your software requirements with the same kind of care, since they're yet another expression of behavior. It's about involving all of your stakeholders in writing acceptance tests.

1. https://dannorth.net/introducing-bdd/